

MC Computers Invoice System Document

[Create Invoice](#) [View Invoices](#)

Generate Customer Invoice

Transaction Date:

06 / 05 / 2025

Products

Product Name:	Quantity:	Unit Price:
Product Name	1	0

[Add Product](#) [Remove](#)

Discount Amount:

0

Subtotal:	\$0.00
Discount:	\$0.00
Total Amount:	\$0.00

[Generate Invoice](#)

Swagger
Supporting SMART BEAR

MCComputers.API 1.0 OAS3

<https://localhost:7133/swagger/v1/swagger.json>

Invoices

POST	/api/Invoices
GET	/api/Invoices
GET	/api/Invoices/{id}

Schemas

- InvoiceCreateDto >
- InvoiceDto >
- InvoiceItemCreateDto >
- InvoiceItemDto >
- ProblemDetails >

MC Computers Invoice System - Quick Start Guide

This guide will help you quickly set up and run the MC Computers Invoice System on your local machine.

Tech Stack

- **Frontend:** Angular (Standalone Components)
- **Backend:** .NET 8 Web API
- **Database:** MS SQL Server
- **Testing (Backend):** xUnit, Moq

Prerequisites

Before you begin, make sure you have the following installed:

- **.NET 8 SDK**
- **Node.js** (LTS version, e.g., v18 or v20) & npm
- **Angular CLI** (install globally: `npm install -g @angular/cli`)
- **SQL Server** (Developer or Express Edition)
- **SQL Server Management Studio** (SSMS) or Visual Studio for database management.
- **Visual Studio 2022** (with ASP.NET and web development workload) OR a code editor like VS Code.

Implemented Features

- Database Connection
- Backend API
- Frontend web (angular)
- Bill create
- Sava bill and detail to the database
- Bill View
- Download Bill asPDF

Setup and Running the Application

Follow these steps to get the application up and running:

1. Backend Setup & Database (MCComputers.API)

1. Clone/Download the Repository.

Git clone <https://github.com/LasaKaru/MC-Computers.git>

2. Configure Database Connection:

- Open the `MCComputers.API` project.
- Edit the `appsettings.json` file (and `appsettings.Development.json`).
- Locate the `ConnectionStrings` section and update `YOUR_SERVER_NAME` with your actual SQL Server instance name (e.g., `localhost`, `(localdb)\mssqllocaldb`, `YOURPCNAME\SQLEXPRESS`):

```
"ConnectionStrings": {  
  "DefaultConnection":  
    "Server=YOUR_SERVER_NAME;Database=MCComputersDB;Trusted_Connection=True;TrustServerCertificate=True;"  
}
```

The `MCComputersDB` database will be automatically created by migrations if it doesn't already exist.

Apply Database Migrations:

- **Using Visual Studio Package Manager Console:**
 - Open `MCComputersApp.sln` in Visual Studio.
 - Go to `Tools > NuGet Package Manager > Package Manager Console`.
 - Ensure `MCComputers.API` is selected as the "Default project".
 - Run the following commands:
 - PowerShell

**Add-Migration InitialCreate
Update-Database**

(If `InitialCreate` already exists, you might only need `Update-Database` or skip `Add-Migration` if no schema changes were made.)

Using .NET CLI:

- Navigate to the `MCComputers.API` project directory in your terminal.
- Run:

```
dotnet ef migrations add InitialCreate  
dotnet ef database update
```

(The same note as above regarding existing migrations applies.)

1. Run the Backend API:

- **Using Visual Studio:**
 - Set `MCComputers.API` as the startup project.
 - Press F5 or click the "Start" button.
- **Using .NET CLI:**
 - In the `MCComputers.API` directory, run: `dotnet run` *The API will start, typically on a port like `https://localhost:7XXX`. Make a note of this HTTPS port. Swagger UI will be available at `/swagger` (e.g., `https://localhost:7123/swagger`).*

2. Test Backend API (Optional - Recommended)

1. Once the API is running, open Swagger UI in your browser (e.g., `https://localhost:7133/swagger`).
2. Expand the `POST /api/Invoices` endpoint.
3. Click "Try it out."
4. Paste the following JSON into the "Request body":

```
{  
  "transactionDate": "2023-10-27T10:00:00Z",  
  "discount": 10.50,  
  "invoiceItems": [  
    {  
      "productName": "Keyboard X1",  
      "quantity": 2,  
      "unitPrice": 25.00  
    },  
    {  
      "productName": "Mouse G5",  
      "quantity": 1,  
      "unitPrice": 40.00  
    }  
  ]  
}
```

}

Invoices

POST /api/Invoices

Parameters Cancel Reset

No parameters

Request body application/json

```
{
  "transactionDate": "2023-10-27T10:00:00Z",
  "discount": 10.50,
  "invoiceItems": [
    {
      "productName": "Keyboard X1",
      "quantity": 2,
      "unitPrice": 25.00
    },
    {
      "productName": "Mouse G5",
      "quantity": 1,
      "unitPrice": 40.00
    }
  ]
}
```

Execute Clear

- Click "Execute."
- You should receive a **201 Created** response.
- You can verify in SSMS (or your preferred DB tool) that data has been added to the **Invoices** and **InvoiceItems** tables in the **MCComputersDB** database.

Sample Response

```
{
  "id": "beb1d631-9b51-4215-9e15-a86afd3e44c9",
  "transactionDate": "2023-10-27T10:00:00Z",
  "discount": 10.5,
  "totalAmount": 79.5,
  "balanceAmount": 79.5,
  "invoiceItems": [
    {
      "id": "fbde57c2-6f71-44b8-857c-f22a3005e3a2",
      "productName": "Keyboard X1",
      "quantity": 2,
      "unitPrice": 25,
      "lineTotal": 50
    },
    {
      "id": "fd3103f7-7259-4421-905a-7db51834f6b0",
      "productName": "Mouse G5",
      "quantity": 1,
      "unitPrice": 40,

```

```

    "lineTotal": 40
  }
]
}

```

Responses

Curl

```

curl -X 'POST' \
  'https://localhost:7133/api/Invoices' \
  -H 'accept: text/plain' \
  -H 'Content-Type: application/json' \
  -d '{
    "transactionDate": "2023-10-27T10:00:00Z",
    "discount": 10.5,
    "invoiceItems": [
      {
        "productName": "Keyboard X1",
        "quantity": 2,
        "unitPrice": 25.00
      },
      {
        "productName": "Mouse G5",
        "quantity": 1,
        "unitPrice": 40.00
      }
    ]
  }'

```

Request URL

https://localhost:7133/api/Invoices

Server response

Code Details

201

Response body

```

{
  "id": "beb1d631-9b51-4215-9e15-a86afd3e44c9",
  "transactionDate": "2023-10-27T10:00:00Z",
  "discount": 10.5,
  "totalAmount": 79.5,
  "balanceAmount": 79.5,
  "invoiceItems": [
    {
      "id": "f8de57c2-6f71-44b8-857c-f22a3005e3a2",
      "productName": "Keyboard X1",
      "quantity": 2,
      "unitPrice": 25,
      "lineTotal": 50
    },
    {
      "id": "fd310347-7259-4421-905a-7db51834f6b0",
      "productName": "Mouse G5",
      "quantity": 1,
      "unitPrice": 40,
      "lineTotal": 40
    }
  ]
}

```

Download

Unit Testing (Backend - xUnit)

This is a requirement.

1. Add a New Test Project:

- In Visual Studio, right-click on your Solution (MCComputersApp) in Solution Explorer.
- Select **Add > New Project....**
- Search for "xUnit Test Project". Select it and click **Next**.
- **Project Name:** MCComputers.API.Tests
- **Location:** Let it default to be within your solution folder.
- Click **Next**.
- **Framework:** .NET 8.0.
- Click **Create**.

2. Add Project Reference:

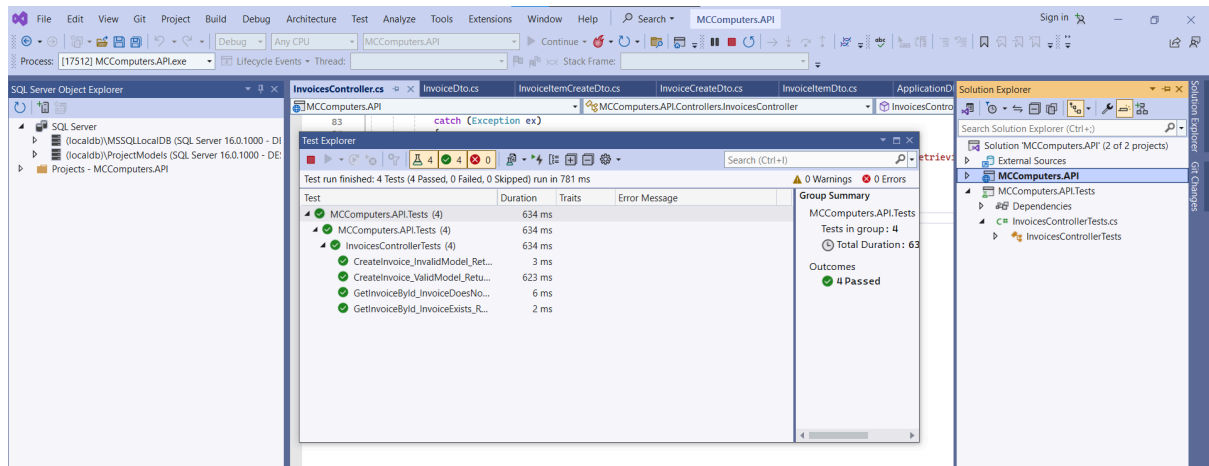
- In the newly created `MCComputers.API.Tests` project, right-click `Dependencies > Add Project Reference....`
- Check `MCComputers.API` and click `OK`. This allows your test project to access the classes in your API project.

3. Install Moq for Mocking:

- Open the Package Manager Console (`Tools > NuGet Package Manager > Package Manager Console`).
- Set `MCComputers.API.Tests` as the "Default project" in the console's dropdown.
- Run the following command:

Install-Package Moq

This package will enable you to create mock objects for dependencies when writing unit tests.



3. Frontend Setup & Run (mc-computers-ui)

1. Navigate to Frontend Directory:

- Open a new terminal or command prompt.
- Change directory to `mc-computers-ui` (e.g., `cd path/to/MCComputersApp/mc-computers-ui`).

2. Install Dependencies:

- Run: `npm install`

3. Verify API URL (Important):

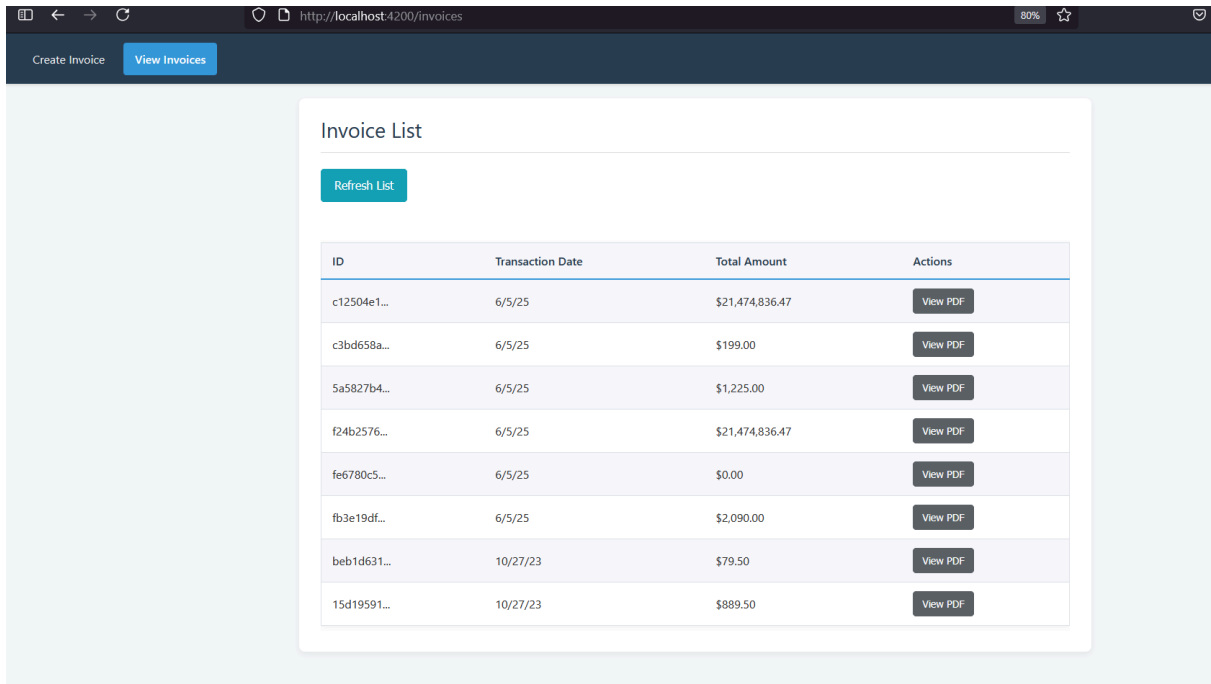
- Open `mc-computers-ui/src/app/services/invoice.service.ts`.
- Ensure the `private apiUrl` variable matches the **HTTPS URL and port** your backend API is running on (from Backend Step 4). Example: `private apiUrl = 'https://localhost:7133/api/invoices'`; (Adjust 7133 if needed).

4. Run the Angular App:

- Run: `ng serve --open` *This will compile the Angular application and open it in your default browser, usually at `http://localhost:4200/`.*

The screenshot shows a web browser at `http://localhost:4200/create-invoice`. The page has a dark blue header with 'Create Invoice' and 'View Invoices' buttons. The main content area is titled 'Generate Customer Invoice' and contains the following form elements:

- Transaction Date:** A date picker showing '06 / 05 / 2025'.
- Products:** A table with three columns: 'Product Name', 'Quantity', and 'Unit Price'. The first row contains 'Product Name', '1', and '0'. A 'Remove' button is next to the unit price.
- Add Product:** A button below the products table.
- Discount Amount:** A text input field showing '0'.
- Summary:** A table showing 'Subtotal: \$0.00', 'Discount: \$0.00', and 'Total Amount: \$0.00'.
- Generate Invoice:** A button at the bottom right of the form.



Invoice

MC Computers
123 Tech Street, Silicon Valley, CA 94000

Invoice ID: 15d19591-42b2-4af7...

Transaction Date: Oct 27, 2023

Product Name	Quantity	Unit Price	Line Total
Mouse G5	10	\$40.00	\$400.00
Keyboard X1	20	\$25.00	\$500.00

Subtotal: \$900.00

Discount: \$10.50

Total Amount: \$889.50

Balance Due: \$889.50

Using the Application

- The application will redirect you to the "Create Invoice" page.

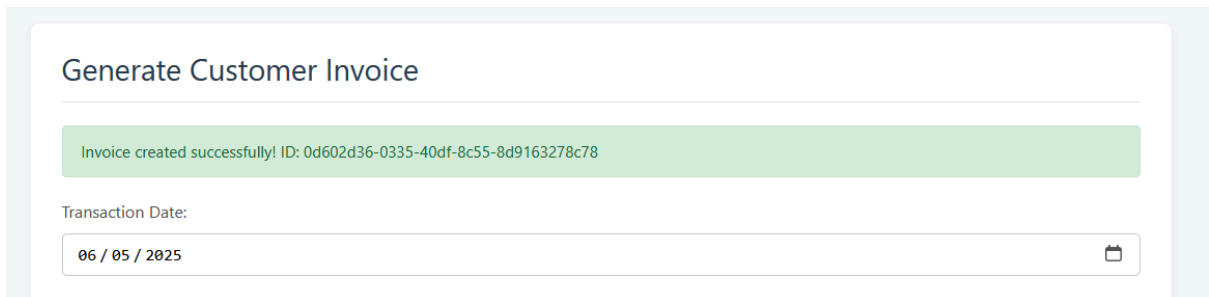
- You can fill out the form, add products, and click "Generate Invoice." This will save the invoice data to the database and automatically generate and download a PDF version of the invoice.
- Use the navigation bar at the top:
 - **Create Invoice:** Takes you to the invoice generation form.
 - **View Invoices:** Shows a list of previously saved invoices. You can click "View PDF" for any invoice in the list to re-generate and download its PDF.

Assumptions

- The migrations will create the necessary database schema; no **.bak** file import is required for this setup.
- You have permissions to create databases and tables on your SQL Server instance (if **MCComputersDB** doesn't exist).
- The default ports for the Angular development server (**4200**) and the .NET API are available.

Make sure to run backend api, otherwise result will be like this.

Successful invoice creation will generate looks like this.



The screenshot shows a web form titled "Generate Customer Invoice". Below the title is a green success message: "Invoice created successfully! ID: 0d602d36-0335-40df-8c55-8d9163278c78". Below this is a label "Transaction Date:" followed by a date input field containing "06 / 05 / 2025" and a calendar icon.

Unsuccessful invoice creation will generate looks like this.

